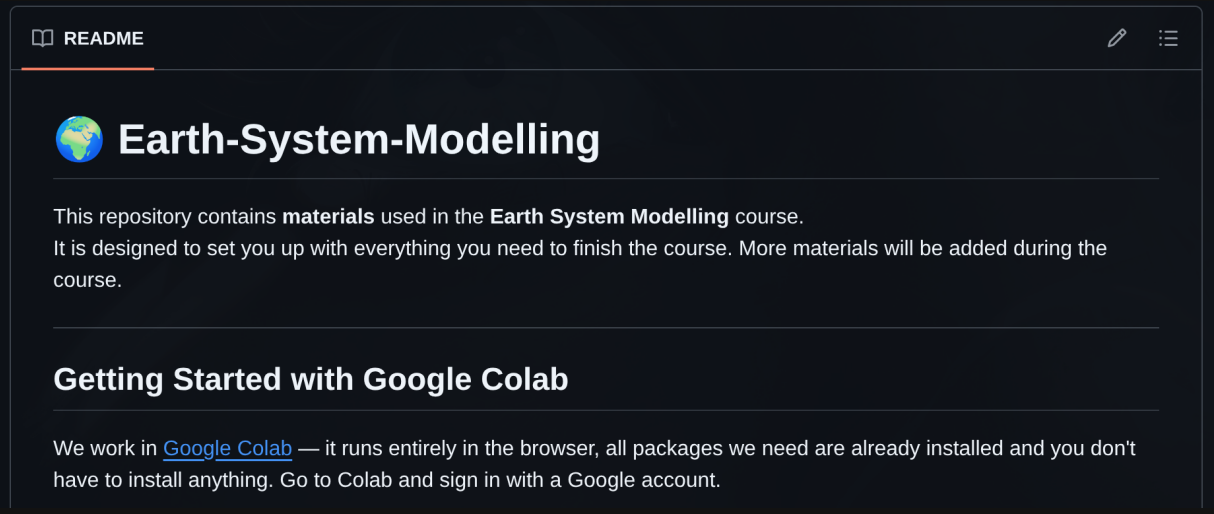
Today

- The GitHub repository
- Python tutorials
- Google Colab
- What to do until the next sessions
- Consultation slots

GitHub repository

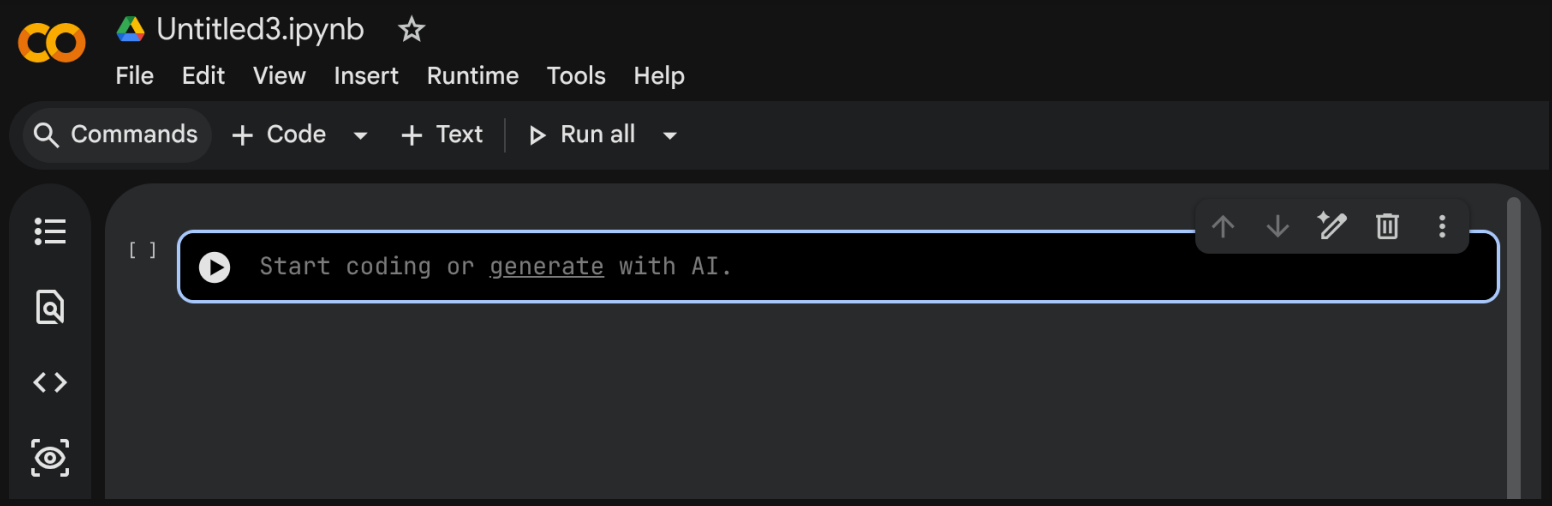
- All course material lives here
- New material is uploaded during the semester
- Data for Germany, Spain and Sweden is in data/

→ github.com/Motzemoere/Earth-System-Modelling



Google Colab

- We work in Google Colab, no installation needed
- Google account necessary
- You can also work locally if you want



Tutorials

- Five tutorials in tutorials/, do them in order
- They cover Python basics and how to build your own models

🌍 Earth-System-Modelling -> Tutorials

Welcome! These tutorials teach you Python from scratch with the goal of building and understanding environmental models like the Soil-Water-Balance Model (SWBM).

Opening Your First Notebook

In Google Colab (recommended):

1. Go to Google Colab and sign in with a Google account.
2. Go to **File → Open notebook → GitHub**.
3. Paste the repository URL `https://github.com/Motzemoere/Earth-System-Modelling` and pick `tutorials/01_Python_Basics_and_Data_Structures.ipynb` from the list.

Collaborating in Colab

- Save your own copy: File → Save a copy in Drive
- Share it with your group via the Share button
- Everyone edits, but each has their own runtime
- Re-run cells to see a teammate’s changes

Example

→ More on teamwork in Colab

```
# Anna runs:
data = pd.read_csv("Data_swbm_Germany.csv")
# Ben then runs the next cell and gets:
NameError: name 'data' is not defined
```

→ Ben has to run Anna's cell first, in their own runtime

A word on AI

- Using AI is not recommended in this course
- AI is here to stay, there is no way around it

Suggestion

- First learn the basics, then learn how to use AI
- You can only check what you understand

Until the next session

- Look at the GitHub repository
- Get familiar with Google Colab
- If you are new to Python, work through the tutorials
- Come to the next seminar (28th October 11am)
- Prepare your questions for the next session

Next session:

- 4th November 11am
- Supervised model implementation

Consultations

- Book your slot by the afternoon of the day before at the latest
- Highly recommended: at least one consultation before Christmas
- Come prepared: concrete questions, your code, what you have tried
- I will not debug your code

→ **Consultation booking table**